Graph Metrics

Let $G = (V, E)$ be an undirected graph, with $n = |V|$ nodes and $m = |E|$ edges. Each edge $e \in E$ has a positive length $\ell(e) > 0$. Shortest-path distances $d(u, v)$ are computed using $\ell(\cdot)$ as weights [12, 6]. When a metric requires connectivity (e.g., radius/diameter/centralities), it is computed on the largest connected component.

The metrics of the graphs associated to road networks discussed in the paper "Planning and Cost Comparison of Optical Access Network Architectures for Distributed Infrastructures" are as follows:

Size and scale

- **Number of nodes (n_nodes):** $n = |V|$. *Counts intersections; problem size.*
- **Number of edges (n_edges):** $m = |E|$. *Counts street or road segments; network size.*
- **Total edge length (total_length):**

$$L_{\text{total}} = \sum_{e \in E} \ell(e). \tag{1}$$

Total physical extent of the network.

Degree-based connectivity Let $\deg(v)$ denote the degree of node v.

- **Average node degree (avg_degree):**

$$\overline{k} = \frac{1}{n} \sum_{v \in V} \deg(v). \tag{2}$$

Typical number of streets per intersection. [12]

- **Number of degree-1 nodes (num_deg1):**

$$n_{\deg=1} = |\{v \in V : \deg(v) = 1\}|. \tag{3}$$

Number of dead ends (leaf nodes).

- **Proportion of degree-1 nodes (prop_deg1):**

$$p_{\deg=1} = \frac{n_{\deg=1}}{n}. \tag{4}$$

Fraction of dead ends; tree-likeness indicator.

- **Number of nodes with degree ≥ 3 (num_deg_ge3):**

$$n_{\deg\geq 3} = |\{v \in V : \deg(v) \geq 3\}|. \tag{5}$$

Number of branching intersections.

- **Proportion of nodes with degree ≥ 3 (prop_deg_ge3):**

$$p_{\deg\geq 3} = \frac{n_{\deg\geq 3}}{n}. \tag{6}$$

Fraction of branching intersections; junction complexity.

- **Degree entropy (degree_entropy):**

$$H_{\deg} = -\sum_{k} p(k) \log p(k), \tag{7}$$

where $p(k)$ is the empirical probability of degree k. *Heterogeneity of node connectivity.* [12]

Density and mesh structure

- **Graph density (density):**

$$\rho = \frac{2m}{n(n-1)}. \tag{8}$$

Edge density relative to a complete graph [12]. For multigraphs (e.g., graphs with pairs of nodes with multiple parallel streets), the edge count includes parallel edges, so this quantity should be interpreted as an edge-density indicator rather than as the classical simple-graph density.

- **Meshedness coefficient / mesh density (mesh_density):**

$$\rho_{\text{mesh}} = \frac{m-n+1}{2n-5}, \qquad (n \geq 3). \tag{9}$$

Planar-oriented meshedness (cyclic redundancy normalized by a planar upper bound), useful for street networks. [2]

Shortest-path extension metrics Eccentricity is defined as:

$$\epsilon(u) = \max_{v \in V} d(u,v). \tag{10}$$

- **Average shortest path length (avg_shortest_path_length):**

$$\overline{d} = \frac{1}{n(n-1)} \sum_{u \neq v} d(u,v). \tag{11}$$

Typical travel distance within the network. [12]

- **Graph radius (radius):**

$$r = \min_{u \in V} \epsilon(u). \tag{12}$$

Best achievable "center" (minimizes the farthest distance). [6]

- **Graph diameter (diameter):**

$$D = \max_{u \in V} \epsilon(u). \tag{13}$$

Maximum shortest-path distance; spatial extent. [6]

- **Average eccentricity (avg_eccentricity):**

$$\overline{\epsilon} = \frac{1}{n} \sum_{u \in V} \epsilon(u). \tag{14}$$

Average remoteness of nodes.

Centrality metrics

- **Average closeness centrality (avg_node_closeness):** Closeness centrality measures how close a node is to all other nodes in terms of shortest-path distance. It is defined as:

$$CC(u) = \frac{n-1}{\sum_{v \neq u} d(u,v)}, \tag{15}$$

where $d(u,v)$ is the shortest-path distance between nodes u and v. In this work, distances are computed using edge lengths, and closeness centrality was computed using the NetworkX implementation [4].

The average closeness centrality is then:

$$\overline{CC} = \frac{1}{n} \sum_{u \in V} CC(u). \tag{16}$$

Higher values indicate nodes that are, on average, closer to all others in the network [8].

- **Maximum betweenness centrality (max_node_betweenness):** Let σ_{st} be the number of shortest paths from node s to node t, and $\sigma_{st}(u)$ the number of those paths that pass through node u. Betweenness centrality is defined as:

$$BC(u) = \sum_{s \neq u \neq t} \frac{\sigma_{st}(u)}{\sigma_{st}}. \tag{17}$$

In this work, shortest paths are computed using edge lengths as weights. The values are normalized following the standard convention so that $BC(u) \in [0, 1]$. The maximum betweenness centrality is:

$$BC_{\max} = \max_{u \in V} BC(u). \tag{18}$$

Nodes with high betweenness act as bottlenecks for shortest-path flow [7, 1].

- **Average betweenness centrality (avg_node_betweenness):**

$$\overline{BC} = \frac{1}{n} \sum_{u \in V} BC(u). \tag{19}$$

Represents the typical importance of nodes for shortest-path flow.

- **Coefficient of variation of betweenness (cv_node_betweenness):**

$$CV_{BC} = \frac{\mathrm{std}(\{BC(u)\})}{\mathrm{mean}(\{BC(u)\})}, \tag{20}$$

defined when the mean is nonzero. *Measures how unevenly centrality is distributed across nodes.*

Branching, cycles, and spanning structure

- **Branching index (branching_index):**

$$B = \frac{n_{\deg=1}}{n_{\deg \geq 3}}. \tag{21}$$

Tree-likeness proxy: many dead ends vs few branching junctions.

- **Cyclomatic number / circuit rank (cyclomatic_number):**

$$\mu = m - n + c, \tag{22}$$

where c is the number of connected components. *Number of "extra" edges beyond a spanning forest; counts independent cycles* [9, 6]

- **Minimum spanning tree total length (mst_length):** Let T be a minimum spanning tree (MST) using $\ell(\cdot)$.

$$L_{\mathrm{MST}} = \sum_{e \in T} \ell(e). \tag{23}$$

Minimum total length to connect all nodes (tree backbone) [11, 15]

- **MST redundancy ratio (mst_ratio):**

$$R_{\mathrm{MST}} = \frac{L_{\mathrm{total}}}{L_{\mathrm{MST}}}, \tag{24}$$

defined when $L_{\mathrm{MST}} > 0$. *Redundancy vs a tree: how much longer the real network is than its MST.*

Edge-length distribution statistics Let $\mathcal{L} = \{\ell(e) : e \in E\}$.

- **Mean edge length (avg_edge_length):**

$$\mu_\ell = \mathrm{mean}(\mathcal{L}). \tag{25}$$

Typical segment length.

- **Median edge length (median_edge_length):**

$$\tilde{\ell} = \mathrm{median}(\mathcal{L}). \tag{26}$$

Robust typical length.

- **Standard deviation of edge length (std_edge_length):**

$$\sigma_\ell = \mathrm{std}(\mathcal{L}). \tag{27}$$

Absolute dispersion of segment lengths.

- **Coefficient of variation of edge length (cv_edge_length):**

$$CV_\ell = \frac{\sigma_\ell}{\mu_\ell}, \tag{28}$$

defined when $\mu_\ell > 0$. *Relative heterogeneity of segment lengths (scale-free)* [10]

- **Minimum edge length (min_edge_length):**

$$\ell_{\min} = \min(\mathcal{L}). \tag{29}$$

Shortest segment present.

- **Maximum edge length (max_edge_length):**

$$\ell_{\max} = \max(\mathcal{L}). \tag{30}$$

Longest segment present.

- **25th percentile of edge length (edge_length_p25):**

$$\ell_{25} = \mathrm{quantile}_{0.25}(\mathcal{L}). \tag{31}$$

Lower-typical segment length.

- **75th percentile of edge length (edge_length_p75):**

$$\ell_{75} = \mathrm{quantile}_{0.75}(\mathcal{L}). \tag{32}$$

Upper-typical segment length.

Community structure

- **Modularity (modularity)** A community is defined as a group of nodes that are more densely connected among themselves than with the rest of the network. In this work, communities are identified using the `greedy_modularity_communities` function from NetworkX [5], which uses Clauset-Newman-Moore greedy modularity maximization [3] to approximate the community partition with the largest modularity. Modularity is defined following [13], and its value is computed using the NetworkX implementation, which follows the standard formulation:

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j), \tag{33}$$

  where A_{ij} is the adjacency matrix, k_i and k_j are node degrees, and $\delta(c_i, c_j)$ equals 1 if nodes i and j belong to the same community, and 0 otherwise. *Higher values indicate stronger community structure.*

- **Number of communities (n_communities).** The number of communities corresponds to the number of groups returned by the `greedy_modularity_communities` function in NetworkX. *It indicates how many distinct modules the network is partitioned into under this optimization criterion.*

Geometric efficiency and route directness

- **Average straightness (avg_straightness_length).** For a pair of nodes (u, v), straightness is defined as the ratio between the Euclidean distance $d_E(u, v)$ and the network shortest-path distance $d_G(u, v)$, following spatial network analysis approaches:

$$S(u, v) = \frac{d_E(u, v)}{d_G(u, v)}. \tag{34}$$

  Let P denote the set of node pairs considered in the computation, and $|P|$ its cardinality. We define the graph-level straightness as the average of this quantity over P:

$$\overline{S} = \frac{1}{|P|} \sum_{(u,v) \in P} \frac{d_E(u, v)}{d_G(u, v)}. \tag{35}$$

  This metric is inspired by the definition of straightness centrality [14], and quantifies how direct routes are compared to straight-line distances.

- **Average tortuosity (avg_tortuosity_length).** Tortuosity is defined as the inverse of straightness for each node pair:

$$T(u, v) = \frac{d_G(u, v)}{d_E(u, v)}. \tag{36}$$

  Using the same set of node pairs P, the graph-level average tortuosity is defined as:

$$\overline{T} = \frac{1}{|P|} \sum_{(u,v) \in P} \frac{d_G(u, v)}{d_E(u, v)}. \tag{37}$$

  This metric measures the detour imposed by the network relative to the Euclidean distance.

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